
Machine-Learning-Guided Tracklet Association and Global Alignment for the FASER Spectrometer

Status report: physical refit chain, geometry-aware Transformers, and the multi-DoF alignment loop

X. Cheng • FASER Alignment Group Meeting

18 August 2026 • all results reproducible from *alignment_ML* artifacts

Outline

What this talk covers

1 Motivation

- Why global alignment, and why association comes first

2 Dataset

- Real physical misalignment banks on MC24, source-disjoint train/validation, sealed test policy

3 Models

- Pairwise MLP + route solver; geometry-aware Transformer V1/V2; V3 structured margin (negative)

4 Results

- Association capture range on translations and rotations
- Single-parameter physical closure: dx and R_y
- Multi-DoF (dx , dy , R_y) iteration-0 closure — decision gate

5 TODO / next steps

- Iteration-1 production (running), truth-free update, coverage, global fitter, more DoF

Reading guide. Every number on these slides is read back from versioned artifacts in outputs/; figure captions name the source file. The permanently sealed test bank is never opened — all model results shown are validation-only unless explicitly labelled “historical sealed test (recorded)”.

Motivation

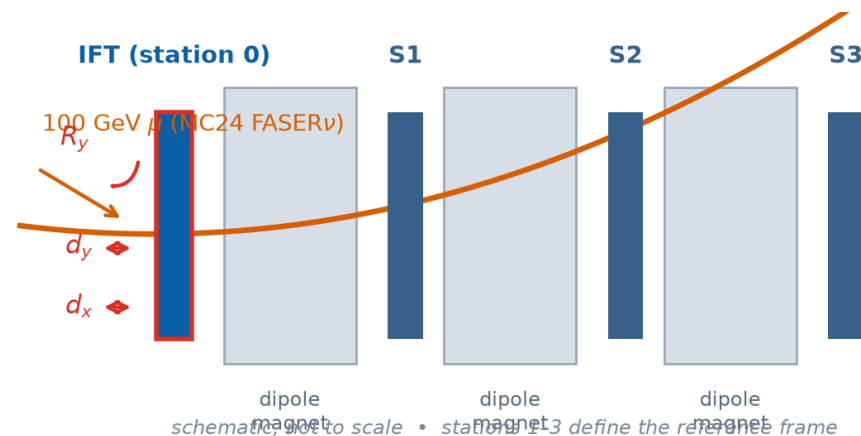
Why this project exists

Physics case

- The FASER spectrometer (IFT + three SCT stations in dipole fields) measures ~100 GeV–TeV muon tracks; station-level misalignment directly biases momentum and decay-vertex reconstruction.
- Alignment needs tracks that **cross all four stations**: a local *tracklet* per station must first be **associated** into a global route before any global fit can run.
- At millimetre-scale misalignment the association itself fails first — you cannot align with tracks you never built.

Why machine learning

- Association is a discrete pattern-recognition problem on a physical candidate graph; learned geometric models may widen the **capture range** beyond χ^2 -only matching.
- An ML backbone plus an iterative *associate* $\rightarrow$ *fit* $\rightarrow$ *align* $\rightarrow$ *refit* loop targets global alignment **without truth association**.



Spectrometer layout and the three active alignment degrees of freedom of the IFT (station 0).

This project's claim to evidence: an end-to-end **real** Calypso/Acts physical chain (no coordinate-shift surrogates), strict source-disjoint train/validation governance, and a sealed test bank with frozen, hash-pinned contracts.

Approach: Real Physical Chain + Iterative Loop

Every misalignment point is an independent re-reconstruction, not a coordinate shift

Per-point physical chain (identical for nominal, anchor, and every finite-difference probe)



Alignment update (current iteration)

- Truth-fixed multi-source **finite-difference (FD) Jacobian** from $\pm$ probe refits; weighted least-squares local step; train fits, **validation is held out**.
- Payload semantics verified against Calypso source: per-station [dx, dy, dz, Rx, Ry, Rz], global T·Rz·Ry·Rx; both segment refit and Acts surfaces follow the payload.
- Local segment q/p is **not** a usable measurement → propagation always mode 0 (enforced at ~25 checkpoints).

Statistical governance

- Splits are **original-xAOD-file disjoint**: 10 train / 8 validation sources; sealed multi-direction test (2 sources, 19 events) frozen 2026-08-12, hash-pinned, read only by two sanctioned historical scans.
- Model selection, calibration, thresholds: **validation-only**. 85 contract JSONs audited: no test access outside the frozen scans.
- Full test suite: **163 passed**.

Dataset

MC24 100 GeV FASERv muons — IFT+1+2+3 control samples, all produced through the real chain

Corpus (outputs/)	Sources	Events	Condition axis	Points / source	Status
Expanded dx/dy physical mc24_v3_expanded_trainval_physical_v1	10 train + 8 val	994 / 796	$dx, dy \in \{0, 0.1, 1, 5, 10, 50\}$ mm	6	complete 108/108
IFT R_y physical mc24_ift_ry_expanded_trainval_physical_v1	10 train + 8 val	994 / 796	$R_y \in \{0, \pm 10, \pm 25, \pm 40, \pm 60\}$ mrad + 4 joint	13	complete 234/234
Multi-DoF iteration-0 anchor bank mc24_multidof_ift_iteration00_anchor_*	10 train + 8 val	100 / source	anchor (+2.0 mm, −1.5 mm, +35 mrad) + FD probes	8	complete 144/144
Multi-DoF joint curriculum bank mc24_multidof_ift_joint_curriculum_*	10 train + 8 val	100 / source	random joint (dx, dy, R_y)	13	complete 234/234
Multi-DoF iteration-1 bank mc24_multidof_ift_iteration01_anchor_*	10 train + 8 val	100 / source	anchor (−0.14 mm, +0.11 mm, +0.74 mrad)	8	in production — cluster 996698
Historical sealed multi-direction test (100116/100117)	2 test	19	$dx/dy \times \{0, 0.1, 1, 5, 10, 50\}$ mm	19	sealed, untouched

- **Tracklet contract** `faser-tracklets-v1`: station ID, global (x, y, z, t_x , t_y), full covariance of [x, y, t_x , t_y], fit quality, hit pattern, MC truth labels.
 - Split unit is the original xAOD file; train/validation UID intersection = 0; test never resolved.
- Six sources hold 98–99 events instead of 100 (input-xAOD property, uniform across points — not job loss).
 - All corpora: q/p mode 0, physical geometry re-propagation, no coordinate surrogates anywhere.

Models

Association models on the existing physical candidate graph — the graph itself is never modified

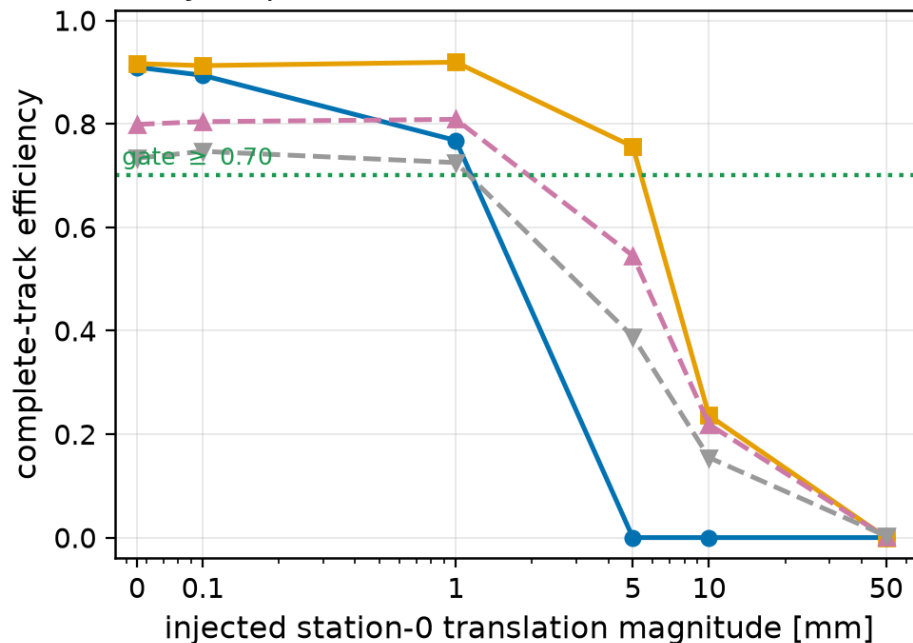
Model	Architecture	Training	Status
Pairwise MLP + route	Edge MLP 12→64→64→1 on residual_v1 features; adjacent unit-capacity route set-packing solver (MILP + DP fast path) with dustbins	Physical curriculum 0/0.1 → 5 → 50 mm; validation-only Platt calibration + thresholds	canonical baseline
V1 geometry-aware Transformer	Tracklet encoder (state, covariance diag, χ^2 /ndf, hits, layer occupancy, station embedding) → 4 sparse blocks, 8 heads, width 128, FFN 128→256→128; attention only on existing physical edges: QK/ $\sqrt{d}$ + $\text{MLP}(g_{ij}) - \lambda \chi^2_{ij}/2\tau$, $g = [r, \text{pulls}, \log_{10} p \chi^2, \log_{10} \det S, \Delta z]$ + pair/direction embeddings; local edge residual head	Weighted BCE, event-graph-uniform sampling, hard-negative x2; curriculum 8/10/12 epochs on top of local 25/30/35	canonical; no capture-range expansion
V2 route-aware	V1 backbone frozen + route query per existing complete chain IFT→S1→S2→S3 (4 node states + 3 edge vectors + pair embeddings + 3 backbone logits); solver-consistent residual utility $L = L_{\text{edge}} + w \cdot (L_{\text{route}} - L_{\text{edge}})$	Edge focal BCE + route-consistency focal BCE + endpoint log-sum-exp competition + fake-endpoint softplus; route bank 800,228 train / 148,551 val routes, truth-blind	frozen association backbone of the loop
V3 structured margin	Loss-augmented structured global-assignment objective	Exact-margin and soft variants on expanded corpus	negative result — suspended

- V1 mechanism diagnosis (1,440 val events, 175k directed message edges): global messages barely change edge ranking (score MAE 0.0135, rank corr 0.9994; edge AP $\Delta \leq 0.003$) while node states over-smooth (mean within-station cosine 0.319 → 0.549) — the **local edge decoder dominates**.
- V2 direct route-probability replacement is solver-inconsistent (no complete route ever selected); the residual form fixed it. Route query itself has validation AP 0.4095 / AUC 0.9449; calibration ECE 0.0177 → 0.00312.

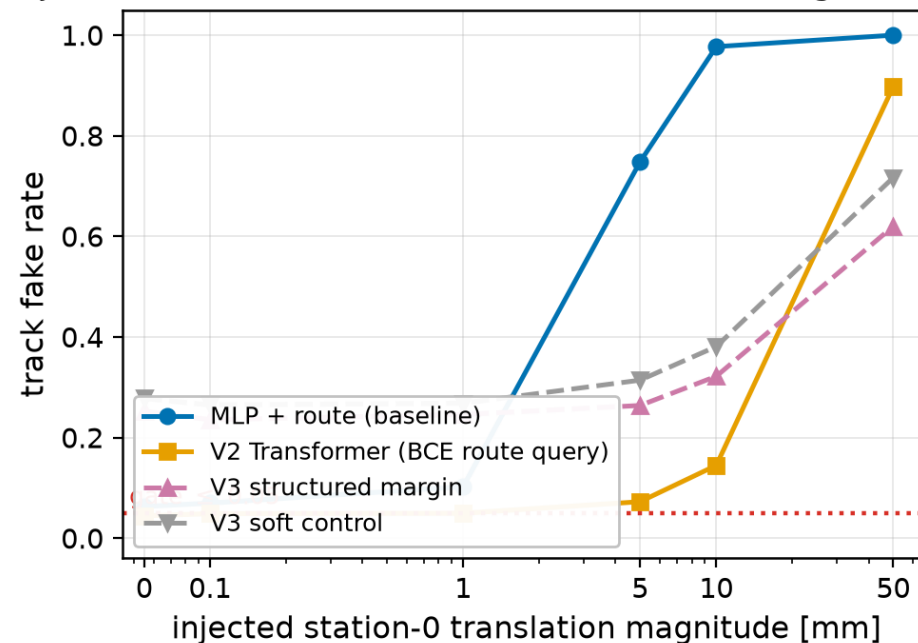
Result A — Association on Large Translations Fails First

Expanded dx/dy corpus, validation-only frozen controls

Expanded dx/dy corpus — validation (994/796 events, source-disjoint), candidate truth-chain recall = 1.0 at all magnitudes



- Raw candidate graph keeps **100% truth-chain recall at every magnitude** — the failure is downstream, in association.
- V2 BCE route query passes the gate at 0–1 mm (nominal 0.916 / 0.967 / 0.044); **every model fails at ≥ 5 mm** (V2 5 mm: eff 0.756, purity 0.946, fake 0.072).

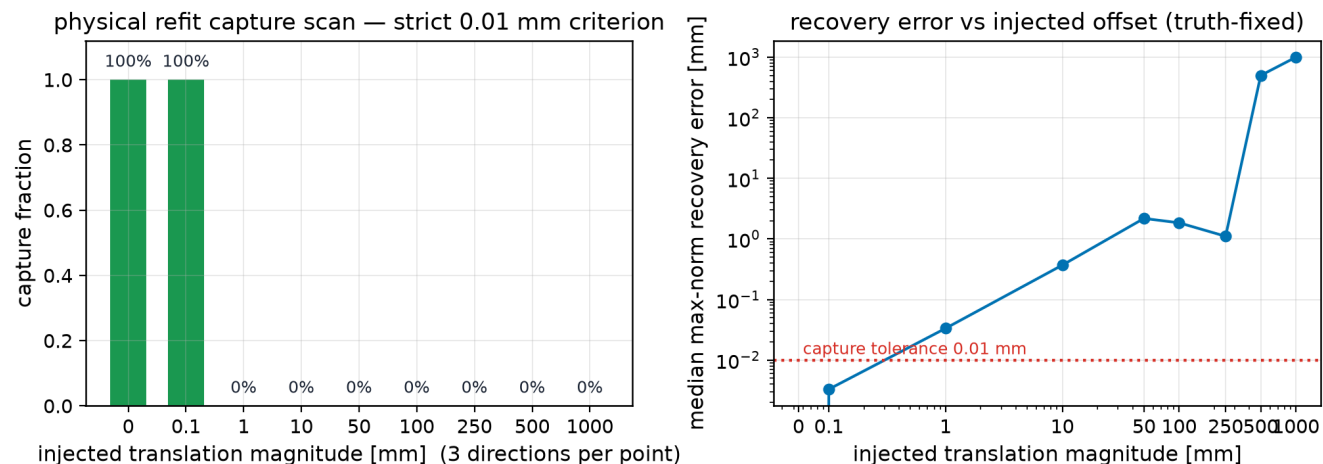


- MLP baseline fails the fake gate even at nominal (0.063).
- V3 structured margin: capture = 0 at all six magnitudes → **confirmed dead end**, suspended.

outputs/mc24_v3_expanded_trainval_validation_control_plots_v3/route_validation_control_summary.csv

Result A' — Physical Refit Capture Boundary (Truth-Fixed)

25-point scan: 9 magnitudes × 3 directions, full re-reconstruction per point



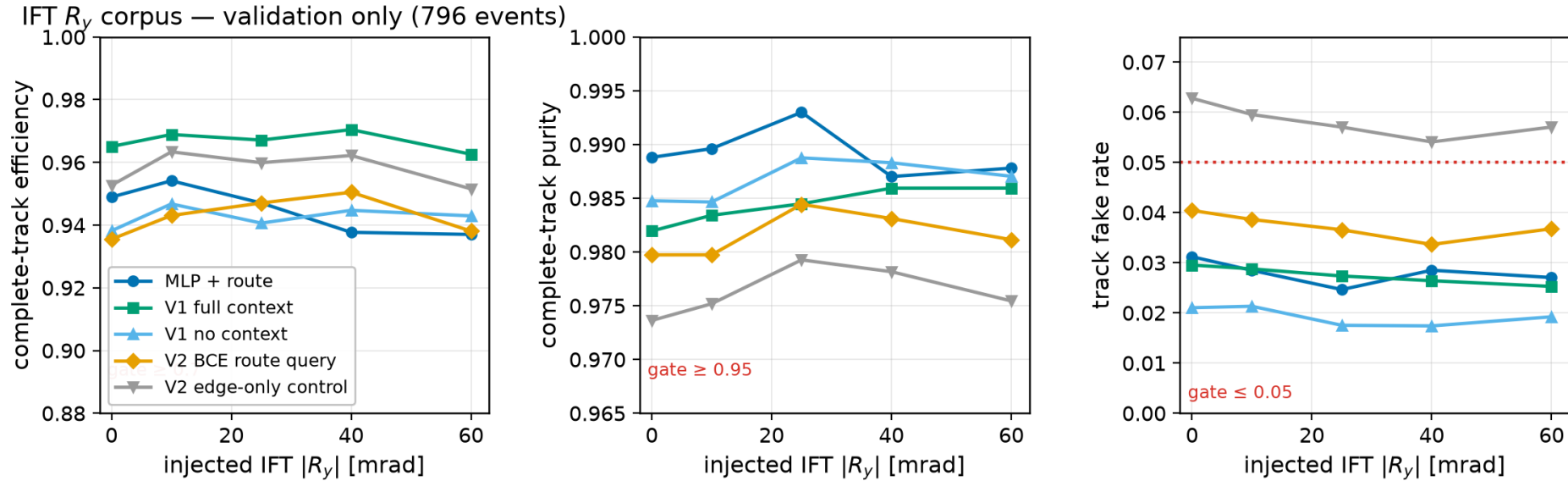
outputs/mc24_muon_fasernu_physical_capture_scan_v1/capture_scan_summary.json

- With truth-fixed pairs, a one-shot linear update on the real refit chain captures **all 0.1 mm trials (3/3)** and **fails all ≥ 1 mm trials (0/3)** under the strict 0.01 mm criterion; ≥ 500 mm loses fit rank entirely.
- Sanity anchor: station-3 +1 mm physical closure recovered to max error **4.9×10^{-13} mm** over 27 truth-matched pairs — the chain itself is exact; capture is limited by linearization + association range.
- Configuration-specific control result (χ^2 gate, seed windows), not a universal detector tolerance.

The 0.1–1 mm association boundary above motivates iterating around a **moving anchor** rather than demanding one giant linear step.

Result B — Rotations Are Not the Association Bottleneck

IFT R_y corpus, validation-only frozen controls (796 events, 8 held-out sources)



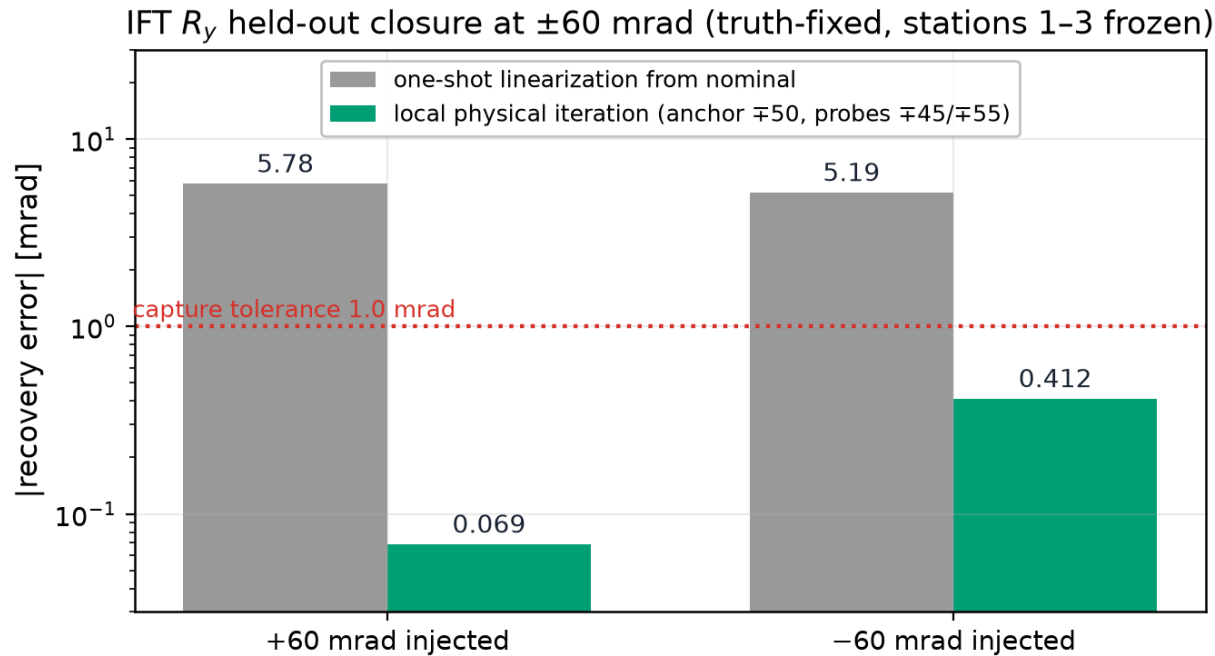
- All frozen models **pass the gate out to $|R_y| = 60$ mrad** (capture 5/5); V1 full-context best (eff 0.965 nominal / 0.963 at 60 mrad). Physics check: ± 60 mrad $\rightarrow \Delta x = \mp 111.5$ mm, $\Delta t_x = \pm 0.060$.

- V2 same-checkpoint edge-only control fails the fake gate (0.063) — the route query is **load-bearing** for V2.

outputs/ift_ry_validation_control_assessment_v2/ift_ry_validation_route_controls.csv

Result C — IFT R_y Recovered to < 1 mrad at ± 60 mrad

Local physical iteration with real /Tracker/Align rotation payloads



outputs/mc24_ift_ry_refinement_smoke_v1/local_step_to_p60/closure.json, .../local_step_to_m60/closure.json

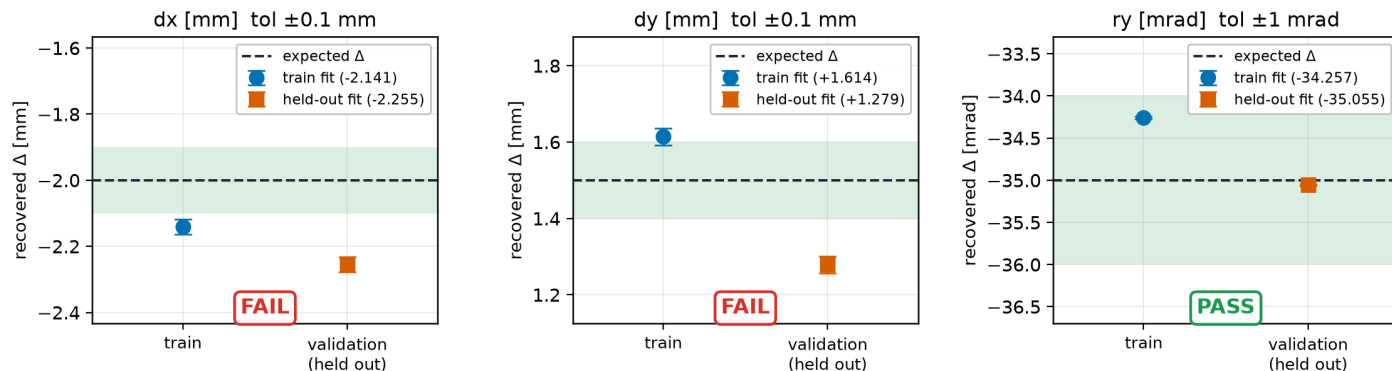
Together with Result B: the rotational leg of the chain is **closed** — identified, recoverable, and not association-limited at 60 mrad.

- **One-shot linearization from nominal fails** at ± 60 mrad (errors 5.78 / 5.19 mrad) — large-rotation response is non-linear.
- **Local iteration closes:** anchor ∓ 50 mrad, probes $\mp 45/\mp 55$ mrad, held-out ± 60 mrad recovered as **+60.069 / -60.412 mrad** (rank 1, condition 1; stations 1–3 at identity to 10^{-15}).
- No truth leakage: derivatives use probe refits only; the held-out point never enters the fit.

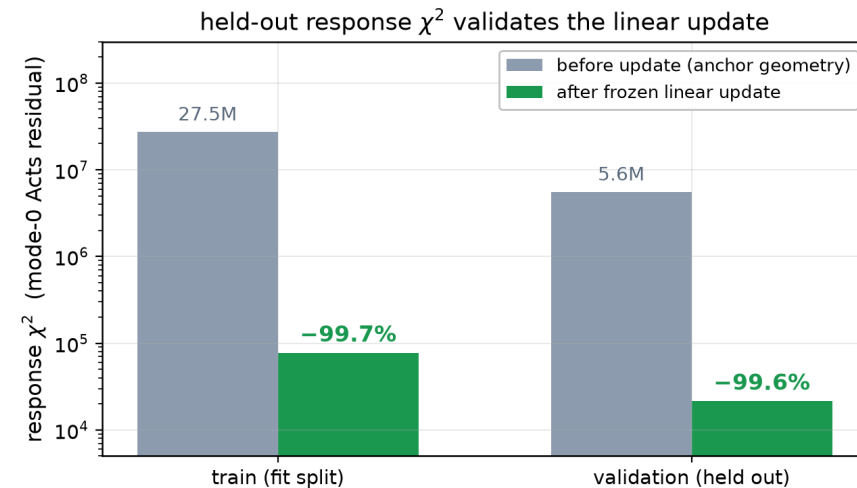
Result D — Multi-DoF Iteration-0 Closure: the Decision Gate

18 sources × 8 real refit points; train determines the update, validation is held out

iteration-0 multi-DoF closure — anchor (dx, dy, R_y) = (+2.0 mm, −1.5 mm, +35 mrad), 2296/1756 obs.



outputs/mc24_multidof_ifit_iteration00_anchor_trainval_closure_v1/multisource_local_step.json

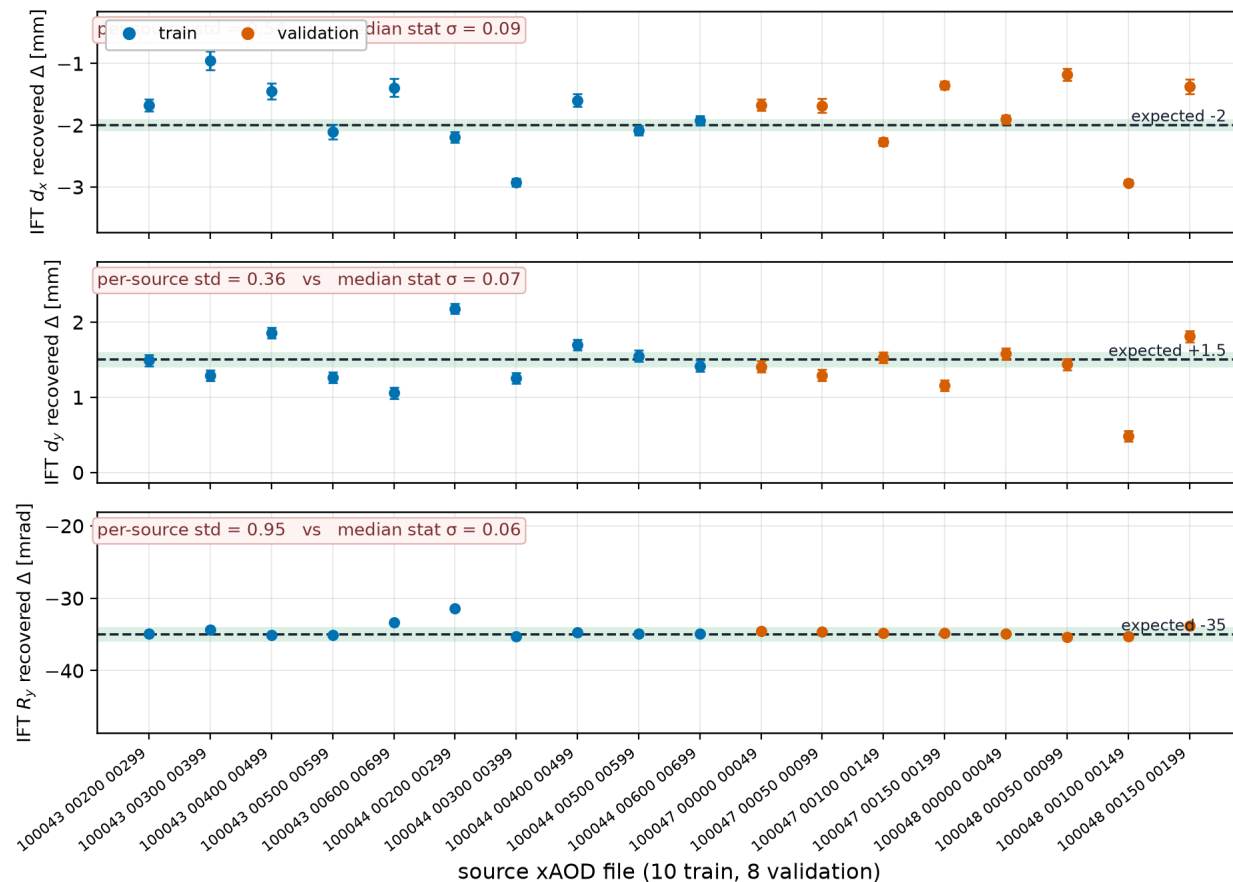


same artifact — response χ^2 application blocks

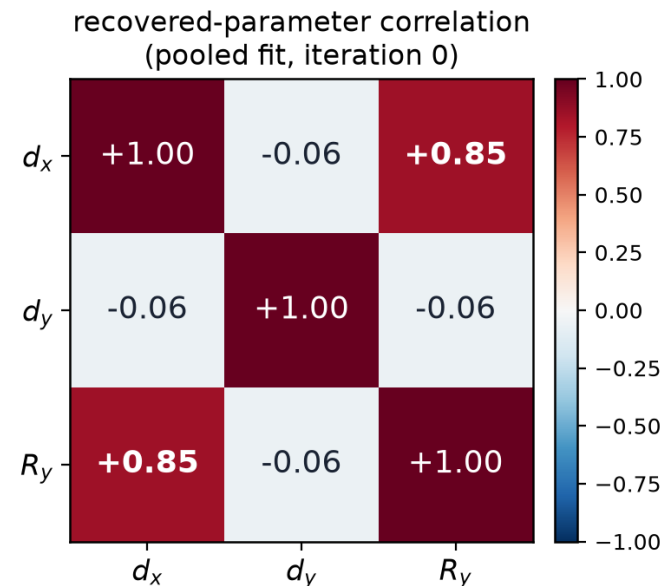
- **Verdict:** `capture_success = false` at frozen tolerances — dx/dy carry 0.11–0.25 mm source-dependent bias ($\sim 6\sigma$); **R_y passes in both splits**; full rank 3, pooled condition 1881.
- Yet the update is **physically right**: held-out response χ^2 drops 99.6%, residuals shrink 10–15×. The earlier 10-event “dx collapse” (6.7 mm error, sign flip) is now shown to be **statistics, not structure** — 100× data brought dx error to 0.14 mm.

Result D' — Where the Remaining Bias Lives

Failure mode changed from statistical collapse to source-dependent systematics



outputs/mc24_multidof_ift_iteration00_anchor_trainval_closure_v1/source_fit_diagnostics.csv (18/18 ok)

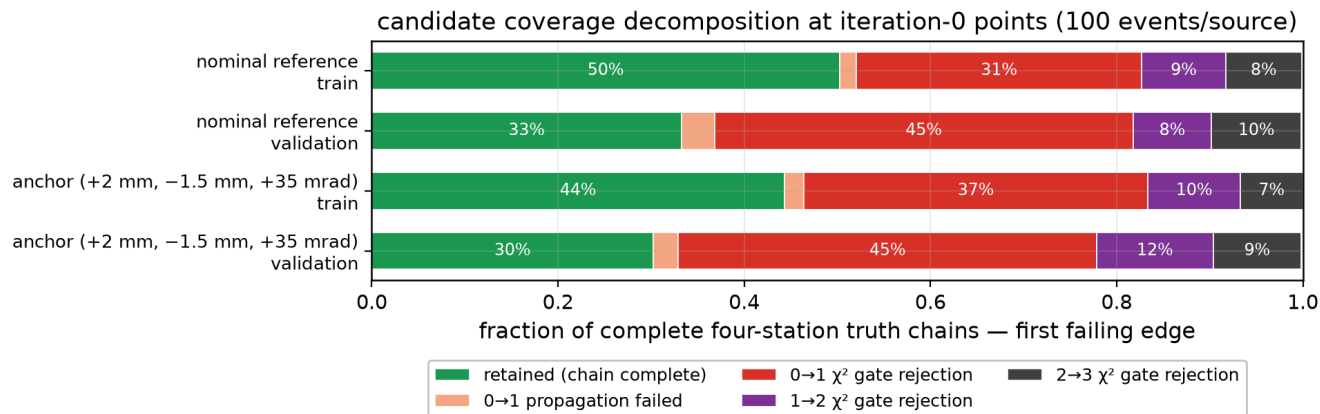


same closure artifact — parameter_correlation

- Per-source scatter is **5-6× the per-source statistical σ** (dx std 0.52/0.54 mm vs $\sigma \approx 0.08-0.10$) — a systematic, not noise.
- Strong **$dx-R_y$ correlation 0.850**: non-linear response at the 35 mrad anchor leaks into dx; $dx-dy$, $dy-R_y \approx -0.06$.
- Per-source bias correlates only weakly with kinematics (max | Pearson | ≈ 0.45) → not a simple phase-space reweighting effect.

Cross-Cutting Bottleneck — Candidate Coverage at the IFT→S1 Edge

First-failure decomposition of complete four-station truth chains ($\chi^2 < 25$ candidate gate)



outputs/mc24_multidof_ift_iteration00_coverage_decomposition_v1/chain_first_failure_by_point.csv

Note: truth-fixed closure above is immune to this; it bounds any future unknown-association update.

- Dominant loss: **0→1 χ^2 -gate rejection**; propagation failures are rare (16–20 chains).
- Retention at anchor: 44% (train) / 30% (validation) vs 50% / 33% at nominal — gates tuned on nominal geometry cut into misaligned candidates.
- Raw *ungated* graph recall on the R_y corpus is $\approx 0.95 \rightarrow$ coverage, not association, sets the ceiling for truth-free alignment updates.

Project Status at a Glance

Eleven working days from environment audit to a running multi-DoF iteration

Date (2026)	Milestone	Evidence
Aug 8–9	Environment + data audit; tracklet export contract; geometry-only χ^2 baseline; MC24 field-aware propagation validated	docs: data_audit, baseline_validation, field_aware_propagation
Aug 9–11	Real payload refit chain (station-3 +1 mm closure to 4.9×10^{-13} mm); 25-point physical capture scan; curriculum MLP baseline	outputs/mc24_muon_fasernu_*capture*
Aug 10–12	Pairwise MLP + global route assignment; historical sealed test frozen; V1 trained + sealed scan (negative); V1 mechanism diagnosis; V2 route-aware validation (w = 0)	docs: geometry_aware_transformer_v1/v2
Aug 13	V3 structured global assignment trained on expanded corpus (994/796) — negative, suspended	docs: structured_assignment_v3
Aug 14	IFT R_y physical rotation: ± 60 mrad closed to < 1 mrad by local iteration; all association controls pass → rotation leg done	docs: ift_ry_physical_rotation
Aug 15–18	Multi-DoF (dx, dy, R_y) loop built; 10-event pilot unstable; 144-point iteration-0 bank complete; closure gate run (this report); project audit + 163 tests green	docs: global_alignment_multidof_loop, project_audit
Aug 18	Iteration-1 bank in Condor production (cluster 996698, 18 jobs × 8 points × 100 events), anchor at the iteration-0 proposal	outputs/mc24_multidof_ift_iteration01_*

TODO — Next Steps in Dependency Order

From the project audit's phased plan; no new ML architecture unless a frozen gate fails with truth chains retained

Immediate (this week)

- **1.** Finish iteration-1 production (running), assemble manifest, rerun aggregated closure at the new anchor.
- **2.** If capture passes at frozen tolerances → first **truth-free** route-selected update with the frozen V2 backbone (`--payload-id inference` → route-selected WLS → verified `--update-json payload`) — the first real iteration of the loop.
- **3.** If the same dx/dy bias persists at the small anchor → attribution studies: per-source Jacobians, FD-step toy scan, R_y -nonlinearity decomposition.

Short term

- **4.** Candidate coverage: understand/relax the $0 \rightarrow 1$ χ^2 -gate loss (anchor retention 44%/30%).
- **5.** Multi-DoF joint-curriculum closure on random joint misalignments (bank complete, manifest refreshed 234/234).

Medium term

- **6.** Field-aware global track fit: needs a Calypso transport-Jacobian export or a validated common-state Acts re-propagation API (interface currently absent — today's update is route-consistency + WLS).
- **7.** Admit R_x , R_z , dz one at a time, each with FD rank/condition/correlation + held-out closure evidence.
- **8.** Station → layer → module hierarchy once the Calypso conditions mapping exists.

Long term

- **9.** Real-data strategy: 2022 IFT re-export failed its provenance check — unresolved.
- **10.** After methodology freeze: build a new source-disjoint multi-DoF test bank, evaluate once. The historical sealed test stays untouched.

Summary

What is established, what is not, and the decision in flight

Established

- End-to-end **real** misalignment chain (payload → refit → Acts) — exact to 10^{-13} mm at truth-fixed closure.
- On translations ≥ 5 mm: **association fails first** while candidate truth recall stays 1.0; no Transformer capture-range expansion shown (honest negative).
- IFT R_y : recoverable to < 1 mrad at ± 60 mrad; rotation is not the association bottleneck.
- Multi-DoF iteration-0: full-rank, validated linear response (held-out χ^2 —99.6%); residual dx/dy bias is source-dependent systematics (5–6 $\times$ stat σ), with dx- R_y correlation 0.85.

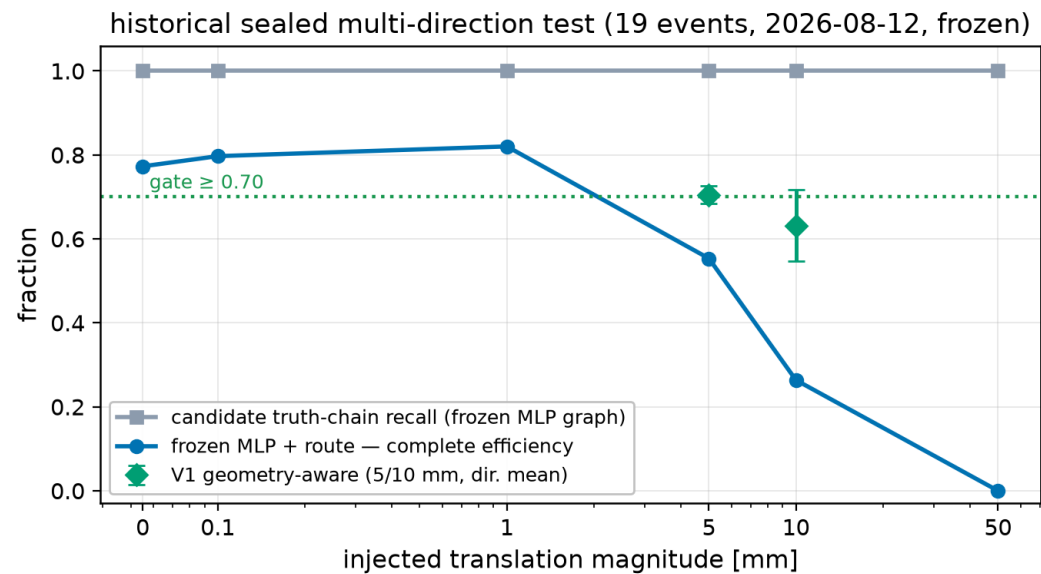
Not yet established

- Multi-DoF joint recovery at frozen tolerances — **iteration-1 closure decides** (in production now).
- Truth-free (unknown-association) alignment update; iterative loop convergence.
- Raw candidate coverage sufficiency (0→1 losses); 6-DoF, layer/module alignment; anything on real data.

Ask of the group: feedback on the iteration protocol (anchor radius, FD steps, tolerances) and on the priority of the 0→1 coverage fix vs. the global-fitter interface.

Backup — Historical Sealed Test & Documented Negative Results

Recorded once, frozen, never re-used for selection



outputs/mc24_muon_2dfluka_multidirection_test_route_level_v1/magnitude_summary.csv (pooled) + V1 sealed scan document

Single frozen 19-event sample — must not be extrapolated; test was never used for model/threshold/calibration selection.

Negative / limiting result	Consequence
V3 structured margin fails the gate at every magnitude (fake 0.25–0.32 at nominal)	line suspended; new hypothesis required
Full-event Transformer context does not widen dx/dy capture range; no-context ablation better at 10 mm	mechanism diagnosis before any larger model
One-shot large-range linearization fails for R_y at ± 60 mrad (5–6 mrad)	local iteration adopted as the protocol
Multi-DoF joint closure statistically unstable at 10 events (dx sign flips)	100 events/source iteration banks
Direct route-probability replacement solver-inconsistent (no complete route selected)	residual utility $L = L_{\text{edge}} + w\Delta L$
Local segment q/p unusable as a V1 measurement	mode-0 propagation enforced everywhere